

NATIONAL UNIVERSITY OF SINGAPORE

NATIONAL UNIVERSITY OF SINGAPORE READING ASSESSMENT 1 FOR Semester 1 AY2023/24

CS1101S Programming Methodology

2023

Time Allowed: 40 minutes

INSTRUCTIONS TO CANDIDATES

1. This question paper contains 19 questions and comprises 10 printed pages, including this page.
2. You are also provided with an answer sheet to write your answers. It comprises 2 printed pages.
3. Use a 2B pencil to write your Student Number in the designated space on the front page of the answer sheet, and shade the corresponding circle completely in the grid for each digit or letter.
DO NOT WRITE YOUR NAME.
4. You must submit only the answer sheet and no other documents.
5. All questions must be answered in the space provided in the answer sheet; no extra sheets will be accepted as answers.
6. For multiple choice questions, use a 2B pencil to shade in the circle of the correct answer completely.
7. Each question has one correct answer. **1 mark** is awarded for each correct answer and there is no penalty for a wrong answer.
8. The full score of this assessment is **18 marks**.
9. Answer all questions.
10. This is a **closed-book** assessment.

1. What is the alphabet letter that is in the box at the top-right corner on the front page of this question paper? Important: Please make sure your answer is correct because it determines how your answers to all the subsequent questions will be graded.

- (a.) A
- (b.) B
- (c.) C
- (d.) D

Section A

In the following questions, the pre-declared `print` function is used in the Python programs. The `print` function prints the value of its input argument to the console. For example, `print(2 * 5)` prints 10.

2. What is the value printed when the following Python \$1 program is evaluated?

```
x = 1

def fun(x, y):
    z = 2
    return lambda y, z: 100 * x + 10 * y + z

print(fun(4, 5)(6, 7))
```

- (a.) Error: one or more names is/are redeclared
- (b.) 147
- (c.) 467
- (d.) 456
- (e.) 156
- (f.) Error: one or more names is/are not declared before being used

3. What is the value printed when the following Python §1 program is evaluated?

```
x = lambda f: f
print(x(1))
```

- (a.) 1
 - (b.) No result: the evaluation leads to an error “f is not a function”
 - (c.) No result: the evaluation leads to an error “x is not a function”
 - (d.) A function that always returns 1
 - (e.) A function that always returns a function
 - (f.) No result: the evaluation leads to an error “1 is not a function”
 - (g.) A function that always returns its argument
4. Consider the following Python §1 program.

```
def f(x):                                # A
    return x

def g(y):
    f = lambda z: z + 1
    return y + f(y)                      # B

def h(w):
    return (lambda f: f)(w)             # C

h(g(f(1)))                               # D
```

Which occurrence(s) of the name `f` is/are in the scope of the declaration in line A?

- (a.) The occurrence in line B, only
- (b.) No occurrences
- (c.) The occurrences in lines C and D, only
- (d.) The occurrences in lines B and D, only
- (e.) The occurrences in lines B and C, only
- (f.) The occurrence in line C, only
- (g.) The occurrence in line D, only
- (h.) The occurrences in lines B, C, and D

5. What is the value printed when the following Python \$1 program is evaluated?

```
f = (lambda x: x)(lambda y: 2)
print(f(1))
```

- (a.) A function that always returns 2
 - (b.) 2
 - (c.) No result: the evaluation leads to an error “2 is not a function”
 - (d.) No result: the evaluation leads to an error “f is not a function”
 - (e.) A function that always returns 1
 - (f.) 1
 - (g.) A function that always returns its argument
 - (h.) A function that always returns a function
6. What is the value printed when the following Python \$1 program is evaluated?

```
f = lambda x: x
g = lambda y: g
h = lambda z: h(z)
print(f(g(h))(1)(2))
```

- (a.) A function that always returns 2
- (b.) No result: the evaluation leads to an error “2 is not a function”
- (c.) 1
- (d.) 2
- (e.) A function that always returns its argument
- (f.) A function that always returns 1
- (g.) No result: the evaluation does not terminate
- (h.) A function that always returns a function

Section B

We know that in Python \$1 the left operand of a binary operation is evaluated before the right operand, and when there are multiple arguments in a function application, the argument expressions are evaluated from left to right.

In some of the following questions, the pre-declared `print` function is used in the programs. The `print` function prints the value of its input argument to the console.

7. What is the sequence of values printed when the following Python 3 program is evaluated?

```
def D(x):  
    print(x)  
    return x  
  
def fun(x, y):  
    return D(D(6 + x) * D(5 + y))  
  
fun(D(4), D(3))
```

- (a.) 4 3 80 10 6 4 8 5 3
 - (b.) 4 3 6 10 5 8 80
 - (c.) None of the other options is the correct answer
 - (d.) 4 3 10 8 80
 - (e.) 4 3 80 6 10 5 8
 - (f.) 4 3 80 10 8
 - (g.) 4 3 6 4 10 5 3 8 80
8. What is the sequence of values printed when the following Python 3 program is evaluated?

```
def fun(n):  
    if n <= 1:  
        return 1  
    else:  
        print(n)  
        return 2 * fun(n - 1)  
  
fun(6)
```

- (a.) 2 3 4 5 6
- (b.) None of the other options is the correct answer
- (c.) 32 16 8 4 2
- (d.) 6 10 8 6 4 2
- (e.) 2 4 8 16 32
- (f.) 6 5 4 3 2
- (g.) 6 12 24 48 96

9. What is the sequence of values printed when the following Python \$1 program is evaluated?

```
def fun(n):  
    if n <= 1:  
        return 1  
    else:  
        x = 2 * fun(n - 1)  
        print(x)  
        return x  
  
fun(6)
```

- (a.) 32 16 8 4 2
 - (b.) 2 4 6 8 10 12
 - (c.) 2 4 8 16 32
 - (d.) 6 5 4 3 2
 - (e.) None of the other options is the correct answer
 - (f.) 2 3 4 5 6
 - (g.) 12 10 8 6 4
10. What is the sequence of values printed when the following Python \$1 program is evaluated?

```
def fun(d, n):  
    print(d)  
    return 1 if n <= 1 else fun(d + 1, n - 1) + fun(d + 1, n - 2)  
  
fun(0, 4)
```

- (a.) None of the other options is the correct answer
- (b.) 0 1 2 3 2 3 1 2 0 1 2 1 2
- (c.) 0 1 2 2 1 2 2 3 3
- (d.) 0 1 1 2 2 3 3 2 2
- (e.) 0 1 2 3 3 2 1 2 2
- (f.) 0 1 1 2 2 2 2 3 3
- (g.) 0 1 2 3 4 5 6 7 8

11. What is the sequence of values printed when the following Python \$1 program is evaluated?

```
def D(m, x):
    print(m)
    return x

def fun(g):
    D("fun", "fun")
    return D("g()", g())

fun(D("arg", lambda: D("4", D("3", 3) + D("1", 1))))
```

- (a.) "3" "1" "4" "arg" "fun" "g()"
- (b.) "arg" "fun" "3" "1" "4" "g()"
- (c.) "fun" "g()" "3" "1" "4" "arg"
- (d.) "arg" "4" "3" "1" "fun" "g()"
- (e.) None of the other options is the correct answer
- (f.) "fun" "g()" "arg" "4" "3" "1"
- (g.) "arg" "4" "fun" "3" "1" "g()"

Section C

12. What kind of process does the following function give rise to for any integer arguments $x > 0$ and $y > 0$?

```
def foo(x, y):
    return (
        1 if y == 0 else foo(x, y - 1)
    ) if x == 0 else foo(x - 1, y)
```

- (a.) An iterative process
- (b.) A process that is neither iterative nor recursive
- (c.) A substitution process
- (d.) A recursive process
- (e.) An infinite process

13. What kind of process does the following function give rise to for any integer argument $n > 1$?

```
def foo(n):  
    return 2 if n == 1 else foo(n - 1) + foo(n - 1)
```

- (a.) An iterative process
 - (b.) An infinite process
 - (c.) A process that is neither iterative nor recursive
 - (d.) A recursive process
 - (e.) A substitution process
14. What kind of process does the following function give rise to for any integer argument $n > 0$?

```
def foo(n):  
    g = lambda x: x - 1  
    return 0 if n == 0 else foo(g(n))
```

- (a.) A substitution process
 - (b.) An iterative process
 - (c.) A recursive process
 - (d.) An infinite process
 - (e.) A process that is neither iterative nor recursive
15. What kind of process does the function `goo` give rise to for any number argument $x > 1$, and any unary function `fun`? Note that `fun` always gives rise to an iterative process for any number argument, and it always terminates.

```
def goo(fun, x):  
    return (  
        fun(x)  
        if x <= 1  
        else fun(goo(fun, x / 2))  
    )
```

- (a.) A recursive process
- (b.) An infinite process
- (c.) A process that is neither iterative nor recursive
- (d.) A substitution process
- (e.) An iterative process

Section D

16. We specify that the function AA, when applied to any three number arguments, should return the smallest value of the three.

```
def AA(x, y, z):  
    return (  
        y if y < z else z  
    ) if x > y else x
```

Which one of the following statements is correct?

- (a.) The function AA meets the specification.
 - (b.) The function AA does not meet the specification because it can be applied to non-positive and/or non-integer argument values.
 - (c.) The function AA does not meet the specification because it does not check whether the arguments are valid inputs.
 - (d.) The function AA does not meet the specification because its returned value is incorrect for some valid argument values.
 - (e.) The function AA does not meet the specification because it is an inefficient way to compute the result.
17. We specify that the function CC, when applied to any two number arguments, should return the sum of the squares of the two input values.

```
def acc(op):  
    return lambda x, y: op(op(y) + op(x))  
  
CC = acc(lambda x: x * x)
```

Which one of the following statements is correct?

- (a.) The function CC does not meet the specification because it causes a run-time error.
- (b.) The function CC meets the specification.
- (c.) The function CC does not meet the specification because it has syntax error(s).
- (d.) The function CC does not meet the specification because it does not return the correct results for all valid argument values.
- (e.) The function CC does not meet the specification because it does not return the correct results for some but not all valid argument values.

18. We specify that the function DD, when applied to any two positive integers, n and p , returns the value of the p th digit from the right in the decimal representation of n . If n has fewer than p digits, DD returns -1.

```
def DD(n, p):  
    return (  
        -1  
        if math_pow(10, p - 1) > n  
        else math_floor(n / math_pow(10, p - 1)) % 10  
    )
```

Which one of the following statements is correct?

- (a.) The function DD does not meet the specification because it can be applied to non-positive argument values.
 - (b.) The function DD does not meet the specification because it does not check if $n > p$.
 - (c.) The function DD meets the specification.
 - (d.) The function DD does not meet the specification because it does not return the correct results for some valid argument value(s).
 - (e.) The function DD does not meet the specification because it has syntax error(s).
19. We specify that the function FF, when applied to any non-negative integer argument n , returns $n!$, the factorial of n .

```
def FF(n):  
    def helper(t, s):  
        return s if t == 0 else helper(t - 1, s + n)  
    return 1 if n == 0 else helper(FF(n - 1), 0)
```

Which one of the following statements is correct?

- (a.) The function FF does not meet the specification because it does not produce the correct results for some valid argument value(s).
- (b.) The function FF meets the specification.
- (c.) The function FF does not meet the specification because it runs into infinite recursion for some valid argument value(s).
- (d.) The function FF does not meet the specification because it is an inefficient way to compute the factorial.
- (e.) The function FF does not meet the specification because it cannot be applied to negative integer argument values.

END OF PAPER